Attachment 2 — Roadmap with hour estimates

> NLnet NGI Zero Commons Fund — Symphact application, 13th open call (deadline 2026-06-01). Condensed from `docs/roadmap-en.md` v1.0. Full text: `roadmap-en.md`.

Summary table

Phase	Milestones	Est. hours	Delivers
Phase 1 Core Runtime	M0.1–M1.0	~148–204	Actor runtime + CFPU reference impl
Phase 2 Kernel Foundation	M2.1–M2.5	~120–160	Scheduler, router, memory, capabilities
Phase 3 Device & I/O	M3.1–M3.5	~80–120	Device actors (UART, GPIO, timer, storage)
Phase 4 Advanced Runtime	M4.1–M4.4	~100–140	Hot code loading, migration, backpressure
Phase 5 Security & Compliance	M5.1–M5.4	~100–140	Capability security, audit, formal verification
Phase 6 Developer Experience	M6.1–M6.5	~100–140	CLI, IDE, profiler, NuGet, docs
Phase 7 First Sellable Product	M7.1–M7.5	~80–120	Reference apps, enterprise, launch
Total		**~728–1024**	**Production-ready Symphact**

> Estimates are calibrated to M0.2 actuals (~8 hours, 580 lines of code, 46 tests, medium complexity).

Phase 1 — Core Runtime (M0.1–M1.0)

Milestone	Status	Tests	Est. hours	Highlights
M0.1 Actor core primitives	✅	30	~6	`IMailbox` / `TMailbox`, `TActorRef`, `TActor<TState>`, `TActorSystem`
M0.2 ActorContext + inter-actor messaging	✅	+16	~8 (actual)	`IActorContext` / `TActorContext`, `DrainAsync` with `MaxRounds`
M0.3 Supervision (let-it-crash)	✅	+30	~24–32	`ISupervisorStrategy`, OneForOne/AllForOne, lifecycle hooks
M0.4 Scheduler + per-actor parallelism	✅	+86	~24–32 (actual ~6h agent team)	`IScheduler`, `TInlineScheduler`, `TDedicatedThreadScheduler`, `QuiesceAsync`
M0.5 Persistence — BCL-only slices	✅	+44	shipped	`IJournal` + `TInMemoryJournal`, `ISnapshotStore` + `TInMemorySnapshotStore`
M0.5 Persistence — content-addressed	🚧	—	~16–24	`TCasJournal` + `TCasSnapshotStore` (SHA-256), supervision lifecycle integration
M0.6 Remoting / Distribution	⏳	—	~28–36	`ITransport` + `TTcpTransport`, serialization, location-transparent `Send`
M0.7 CFPU Hardware Integration	⏳	—	~30–50	`TMmioMailbox`, end-to-end demo over `FenySoft.CilCpu.Sim`
M1.0 Samples + Stabilization	⏳	—	~12–16	CounterActor, ChatRoom, API reference, benchmarks

Total green tests at grant submission: 186 (120 Core + 22 Platform.DotNet + 44 Persistence).

Phase 2 — Kernel Foundation (M2.1–M2.5, ~120–160h)

Milestone	Est. hours	Description
M2.1 Root Supervisor + Actor Hierarchy	~20–28	`TRootSupervisor`, boot sequence, fixed restart strategy
M2.2 Scheduler Actor	~28–36	`ISchedulerPolicy`, `TRoundRobinPolicy`, `TLoadBalancePolicy`, actor-to-core affinity, watchdog
M2.3 Router Actor	~24–32	`TRouter`, location-transparent `Send`, migration support, dead-letter handling
M2.4 Memory Manager	~20–28	`TMemoryManager`, per-core SRAM budget, per-actor GC, memory pressure notification
M2.5 Capability Registry	~28–36	`TCapabilityRegistry`, CST HW table management, delegation + revocation, 8-bit permission flag

Phase 3 — Device & I/O Layer (M3.1–M3.5, ~80–120h)

Milestone	Est. hours	Description
M3.1 Device Actor Framework	~20–28	`IDeviceActor`, MMIO ownership, interrupt → message conversion
M3.2 UART Device Actor	~12–16	TX/RX actor pair, subscribe pattern, backpressure
M3.3 GPIO Device Actor	~8–12	Pin config (Input/Output/Interrupt), edge-triggered IRQ → message
M3.4 Timer / Clock Service	~12–16	`TTimerService`, `AskWithTimeout`, mock clock for testing
M3.5 Storage Service	~28–36	`IStorageProvider`, `TFileSystemActor`, file handle = actor

Phase 4 — Advanced Runtime (M4.1–M4.4, ~100–140h)

Milestone	Est. hours	Description
M4.1 Hot Code Loading	~32–40	`THotCodeLoader`, opcode whitelist, Erlang-style version switch, state preservation
M4.2 Actor Migration	~24–32	Suspend → serialize → move; router mapping update; cross-core / cross-node
M4.3 Backpressure + Flow Control	~20–28	Mailbox depth limit, `TrySend`, backpressure propagation
M4.4 Priority Mailbox + System Messages	~24–32	`TPriorityMailbox`, opt-in per actor, watchdog kill, graceful shutdown

Phase 5 — Security & Compliance (M5.1–M5.4, ~100–140h)

Milestone	Est. hours	Description
M5.1 Full Capability Model	~28–36	CST HW lookup on every Send, audit trail, AuthCode integration
M5.2 Capability Delegation & Revocation	~20–28	Attenuation on delegation, revocation broadcast
M5.3 Audit & Compliance Logging	~20–28	Tamper-proof audit log actor, IEC 61508 / ISO 26262 baseline reporting
M5.4 Formal Verification Foundations	~32–48	TLA+ / Dafny spec of `send` / `receive` semantics, capability invariants

Phase 6 — Developer Experience (M6.1–M6.5, ~100–140h)

Milestone	Est. hours	Description
M6.1 CLI Tooling	~20–28	`symphact` CLI: `new`, `build`, `run`, `deploy`, `monitor`
M6.2 IDE Integration	~24–32	VS Code extension, Rider plugin, live actor tree, message breakpoint
M6.3 Profiler & Monitoring Dashboard	~24–32	Per-actor msg/sec, latency histogram, Prometheus + Grafana export
M6.4 NuGet Packages & API Reference	~16–24	`Symphact.Core`, `Symphact.Supervision`, `Symphact.Persistence`, `Symphact.Remoting`, `Symphact.Security`, `Symphact.Devices`
M6.5 Documentation & Tutorials	~16–24	Getting started, architecture deep-dive, Akka.NET / Orleans migration guide

Phase 7 — First Sellable Product (M7.1–M7.5, ~80–120h)

Milestone	Est. hours	Description
M7.1 Reference Applications	~24–32	IoT Gateway, AI Agent Cluster, SNN Demo, Chat Server
M7.2 Production Hardening	~20–28	10K+ actor stress test, chaos testing, soak test, performance regression CI
M7.3 Enterprise Features	~16–24	RBAC, multi-tenant isolation, configuration management, K8s health check
M7.4 Licensing & Packaging	~8–12	Dual license, Docker image, Helm chart, landing page
M7.5 Launch Preparation	~12–16	Announcement blog, conference talks, GitHub Sponsors, first partner onboarding

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Grant-funded scope (12 months, €30 000)

The grant funds the **next stretch beyond what has shipped** — the remaining M0.5 work, M0.6, M0.7, plus selected pieces of Phase 5 (formal-verification foundations) and Phase 6 (NuGet, docs, outreach). Hot code loading (M4.1) is **explicitly deferred** to a follow-up grant.

Grant milestone	Roadmap mapping	Hours	Amount	Timeframe
M1: Persistence	M0.5 (content-addressed)	~80h	€2 900	Months 1-3
M2: Remoting + Capability Registry	M0.6 + M2.5	~160h	€5 800	Months 2-7
M3: Kernel + Device actors	M2.1, M2.3, M3.1-M3.4	~180h	€6 500	Months 5-10
M4: CFPU integration demo	M0.7 (partial)	~80h	€2 900	Months 8-11
M5: Dev experience + docs + outreach	M6.4 + M6.5 (partial)	~180h	€6 500	Continuous
M6: Formal-verification foundations	M5.4 (partial)	~80h	€2 900	Months 9-12
Engineering hours total		**~760h**	**€27 500**	**12 months**
Online outreach + documentation	(video production, GitHub Pages, doc expansion)	—	€2 500	Continuous
Requested amount			**€30 000**	

Expected `osreq-to-cfpu` issues: 3-7 concrete HW requirements documented over the 12 months — feedback paths include SHA-256/BLAKE3 hash instruction, CST cache, mailbox FIFO depth, supervision fault-notification primitives, SRAM-to-SRAM DMA.